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Research Article

Stable Pacific deep circulation punctuated by episodic intensification during the Mid-Pleistocene Transition

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ABSTRACT

The Mid-Pleistocene Transition (MPT, 600–1300 ka) marks the shift of the dominant ice age cycle from 41-kyr to 100-kyr in the late Pleistocene. While the Atlantic Ocean circulation changes have been found to be closely linked with the MPT, the Pacific's role remains poorly constrained. Here we present fish teeth neodymium isotope (ε_{Nd}) records from three sites along the Pacific Deep Western Boundary Current (DWBC) to capture the Pacific deep ocean circulation changes on Glacial-Interglacial (G-IG) timescale during the MPT. We found reduced deep water ε_{Nd} gradient ($\Delta\varepsilon_{Nd}$) between north and south Pacific during glacial periods. After constraining non-conservative influences on ε_{Nd}, we infer that enhanced northward expansion of Southern Component Water (SCW) and/or shoaling of the Pacific Deep Water (PDW)-Lower Circumpolar Deep Water (LCDW) interface best explain the observed patterns. While the G-IG variability of the $\Delta\varepsilon_{Nd}$ is largely consistent throughout the MPT, indicating a generally stable rhythm of Pacific deep overturning circulation, the pronounced ε_{Nd} values of the three Pacific sites during Marine Isotope Stage (MIS) 22 suggest a transient circulation anomaly. The rigorous Pacific deep ocean circulation enhanced the ocean carbon storage through surface disequilibrium during glacials. Our results indicate that ocean circulation has played an important role in shaping the late Pleistocene ice ages through interaction with carbon cycle.

1. Introduction

The Pleistocene G-IG cycles in climate change have been controlled by Earth's orbital geometry modulated insolation change (Tzedakis et al., 2017). During the MPT, the dominant G-IG cycle shifted from ~41 kyr to ~100 kyr, but it is not seen in insolation change (Berends et al., 2021). Mechanisms triggering the MPT have been ascribed to ice-climate feedbacks, ice sheet basal conditions, ice-dynamical instabilities, sea ice change and atmospheric CO_2 decline (Berends et al., 2021) as well as Antarctic ice sheet growth (An et al., 2024). A prevalent "crisis" narrative (Pena and Goldstein, 2014) has argued that a dramatic weakening of deep ocean circulation perturbed the global carbon cycle (Farmer et al., 2019) and caused the MPT. These changes were most pronounced during MISs 22–25, with MIS 23 being a "failed interglacial" at glacial level, that seems to indicate an ocean circulation crisis during MPT. However, Hines et al. (2024) recently suggested only modest changes in Atlantic circulation across the MPT, challenging the "crisis" narrative. Most records of ocean circulation change across the MPT are based on Atlantic, Indian, and Southern Oceans (Farmer et al., 2019; Pena and Goldstein, 2014; Williams et al., 2024), with limited data from the North Pacific, the largest reservoir of respired carbon, leaving a gap in the global perspective.

However, the Pacific circulation system is particularly complex, characterized by the interaction of multiple deep-water masses and their progressive modification during transport (Fig. 1, Fig. 2). This complexity highlights the need to examine circulation along the Pacific Deep Western Boundary Current (DWBC), which provides the dominant pathway for south-to-north deep-water transport. By establishing a meridional transect of sites along the DWBC, it becomes possible to capture both the source properties of SCW and the upwelling and mixing during northward advection. Variations in DWBC strength and geometry critically regulate the balance between carbon sequestration and release: slower overturning favors the accumulation of respired carbon in the abyss, while more vigorous circulation enhances its redistribution

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and ventilation. Establishing a meridional transect of ε_{Nd} records along the DWBC therefore provides a robust framework to track both water-mass transformation and the carbon storage capacity of the deep Pacific across G-IG cycles.

Neodymium isotopes have long been applied as a quasi-conservative tracer of water masses, reflecting their source signatures and subsequent mixing histories. The Nd isotope ratio $^{143}Nd/^{144}Nd$, expressed as ε_{Nd} (DePaolo and Wasserburg, 1976), varies among modern water masses. Previous studies based on modelling reveal that the residence time of Nd in seawater is 856 years (Robinson et al., 2021), comparable to the rate of global meridional ocean circulation (Broecker and Peng, 1982). ε_{Nd} was traditionally treated as a conservative water mass tracer and widely used to study ocean circulation (Farmer et al., 2019; Hines et al., 2024; Jaume-Seguí et al., 2020; Kim et al., 2021; Pena and Goldstein, 2014; Tachikawa et al., 2021; Williams et al., 2024). In the Pacific, ε_{Nd} provides a valuable tool to reflect advection from the Southern Ocean and progressive modification along the DWBC pathway. Along the modern Pacific DWBC, the major water masses exhibit distinct ε_{Nd} signatures. LCDW is the least radiogenic, ranging from −8 to −5. Upper Circumpolar Deep Water (UCDW) is slightly more radiogenic, varying from −7.5 to −4. PDW has the most radiogenic signature, changing between −4.5 and −3 (Fig. 1B).

However, as seawater ε_{Nd} is getting more radiogenic along the Pacific DWBC, non-circulation factors also have effect on changing seawater ε_{Nd}. The non-conservative behavior of seawater ε_{Nd} mainly results from interactions with sediments, rocks, and particles in the water column. As Global Neodymium Ocean Model (GNOM) results indicate, non-conservative processes could induce a 1–2 ε radiogenic shift to the Pacific DWBC seawater, primarily affecting depths greater than 3000 m (Fig. 3, Pasquier et al., 2022), though among these processes benthic flux to the deep ocean may be underestimated in GNOM. Benthic flux of volcanogenic and continental materials, and water column reversible scavenging dominate non-conservative behavior of the DWBC region, whereas hydrothermal input, submarine groundwater discharge, and riverine input are negligible (Pasquier et al., 2022; Pöppelmeier et al., 2022).

Benthic flux, the release of Nd from sediments and pore waters, is the major source of Nd in ocean bottom waters (Abbott et al., 2015), accounting for an estimated 30–50 % of the global oceanic Nd input (Pasquier et al., 2022; Pöppelmeier et al., 2022). The flux ranges from 3 to 50 pmol cm^{-2} $year^{-1}$, and its ε_{Nd} largely resembles sedimentary terrigenous detritus (Abbott et al., 2015). Continental sediments, with ε_{Nd} ranging from −8 to −15, dominate the Pacific but contribute little to seawater ε_{Nd} due to their low reactivity. Reactive volcanic ash, however, efficiently releases Nd and acts as a significant benthic Nd source in the Pacific (Robinson et al., 2021).

East Asian dust is another important source in the North Pacific. Model simulations suggest that dust strongly affects marginal seas but has a weaker impact in the open ocean (Pasquier et al., 2022; Pöppelmeier et al., 2022). It primarily affects the 20°-50° N region, with ε_{Nd} values ranging from −8 to −15 (Chen et al., 2007; Shao and Dong, 2006). Dust flux also significantly contributes to reversible scavenging, which affects the vertical distribution of Nd concentrations ([Nd]) and ε_{Nd} in seawater. In the North Pacific, this process can transfer surface radiogenic ε_{Nd} signatures to deeper waters (Amakawa et al., 2009).

The non-conservative effect was evident. For example, at ODP 1123, authigenic ε_{Nd} values are consistently ~1 unit more radiogenic than the expected deep seawater signal (Noble et al., 2013). This offset is most likely caused by boundary exchange with radiogenic sediments and volcanic inputs from the Chatham Rise region, while reversible scavenging processes may further transfer radiogenic signatures from shallower waters into the deep ocean, together modifying the local seawater ε_{Nd} composition (Noble et al., 2013). It may also arise from the presence of different sub-water masses carrying distinct Nd isotopic compositions that mix in this region (Basak et al., 2015).

During preservation of marine authigenic materials, such as fossil fish teeth (FFT), past seawater ε_{Nd} signal could also be biased by pore waters (Abbott et al., 2022). FFT, composed mainly of (hydroxy-)

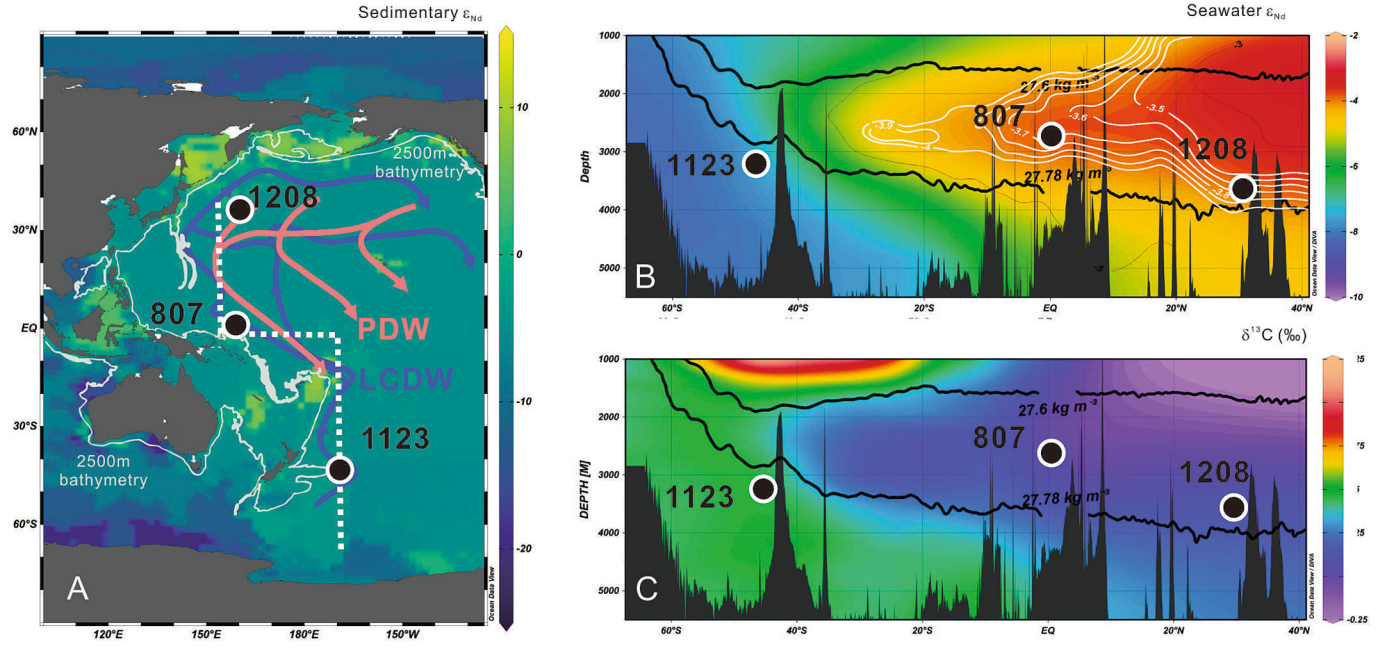


Fig. 1. (A) Site map. The black transect corresponds to the sections in (B) and (C). Blue and red curves with arrows represent schematic flow paths of LCDW and PDW, respectively (Kawabe and Fujio, 2010; Talley, 2013). Light grey contours denote 2500 m bathymetry. Background shading indicates estimated detrital ε_{Nd} of surface sediments (Robinson et al., 2021). (B) ε_{Nd} along the Pacific DWBC section (black solid line in A). Circles indicate observational data (GEOTRACES Intermediate Data Product Group, 2023; van de Flierdt et al., 2016); background shading shows model output, which misfits in details and please refer to Fig. 5/6/7 for actual seawater data (Pasquier et al., 2022). (C) $\delta^{13}C$ along the Pacific DWBC section (white dashed line in A) (Lauvset et al., 2022). Black contours in (B) and (C) represent isopycnals of 27.6 and 27.78 kg m^{-3}. White contours in (B) denote the modeled ε_{Nd}. Figures created using Ocean Data View Software (http://odv.awi.de). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

fluorapatite, adsorb Nd and other REEs as they fossilize at the sediment surface (Martin and Scher, 2004). As shown in the plots of paired authigenic-detrital ε_{Nd} measurements from the modern or core-top samples, FFT displays the largest deviation from the theoretical 1:1 relationship line (Fig. 4). This indicates that the FFT ε_{Nd} has the minimum detrital overprint and should be the best proxy of ocean circulation change relative to foraminifer and Fe-Mn leachates.

Most existing studies of the MPT have focused on the Atlantic, leaving a major gap in understanding Pacific deep circulation, despite the basin's dominant role in carbon storage. At the same time, efforts to reconstruct circulation face two key constraints: the uncertainties in applying Nd isotopes as a quasi-conservative tracer, and the inherent complexity of Pacific water-mass structure. To overcome these limitations, we developed a transect approach along the Pacific DWBC. We generated three new FFT ε_{Nd} records from ODP Sites 807, 1208, and 1123, spanning 600–1300 ka, and integrated them with bulk sediment ε_{Nd} analyses to assess potential detrital influence. To further disentangle circulation signals from non-conservative influences, we employed the GNOM to evaluate the role of non-conservative processes such as benthic flux, dust, and volcanic input. By combining proxy records with model-based constraints, our approach aims to resolve G-IG variations in Pacific deep overturning and clarify their contribution to ocean carbon storage and the MPT.

2. Material and methods

2.1. Oceanography, site information and age model

The Pacific DWBC transports deep waters formed in the Southern Ocean, primarily LCDW into the north Pacific. The Pacific DWBC flows along the east side of the Campbell Plateau and Kermadec Trench, acting as a conduit through which Antarctic Bottom Water (AABW) and LCDW is transported away from the Antarctic (Chandler et al., 2024). The northwardly flowing LCDW fills the deep Pacific and upwells to the mid-depth in the mid-latitude North Pacific to form PDW. The southwardly flowing PDW overlies LCDW and mixes with UCDW to form modified PDW, which re-enters the Antarctic Circumpolar Current (ACC) through the east Pacific (Talley, 2013; Talley, 2008).

LCDW below 3000–3500 m has densities >27.78 kg m^{-3} (Talley, 2013; Reagan et al., 2024). During the northward flowing, LCDW progressively mixes with PDW, with a gradual decrease in $[O_2]$ (Fig. 2B, Lauvset et al., 2022) and an increase in ε_{Nd} (Fig. 1B). The density of PDW between ~1500 m and 3000 m varies from 27.6 to 27.78 kg m^{-3} (Fig. 1B). The core of PDW has an ε_{Nd} of ~−3.5, and as it flows

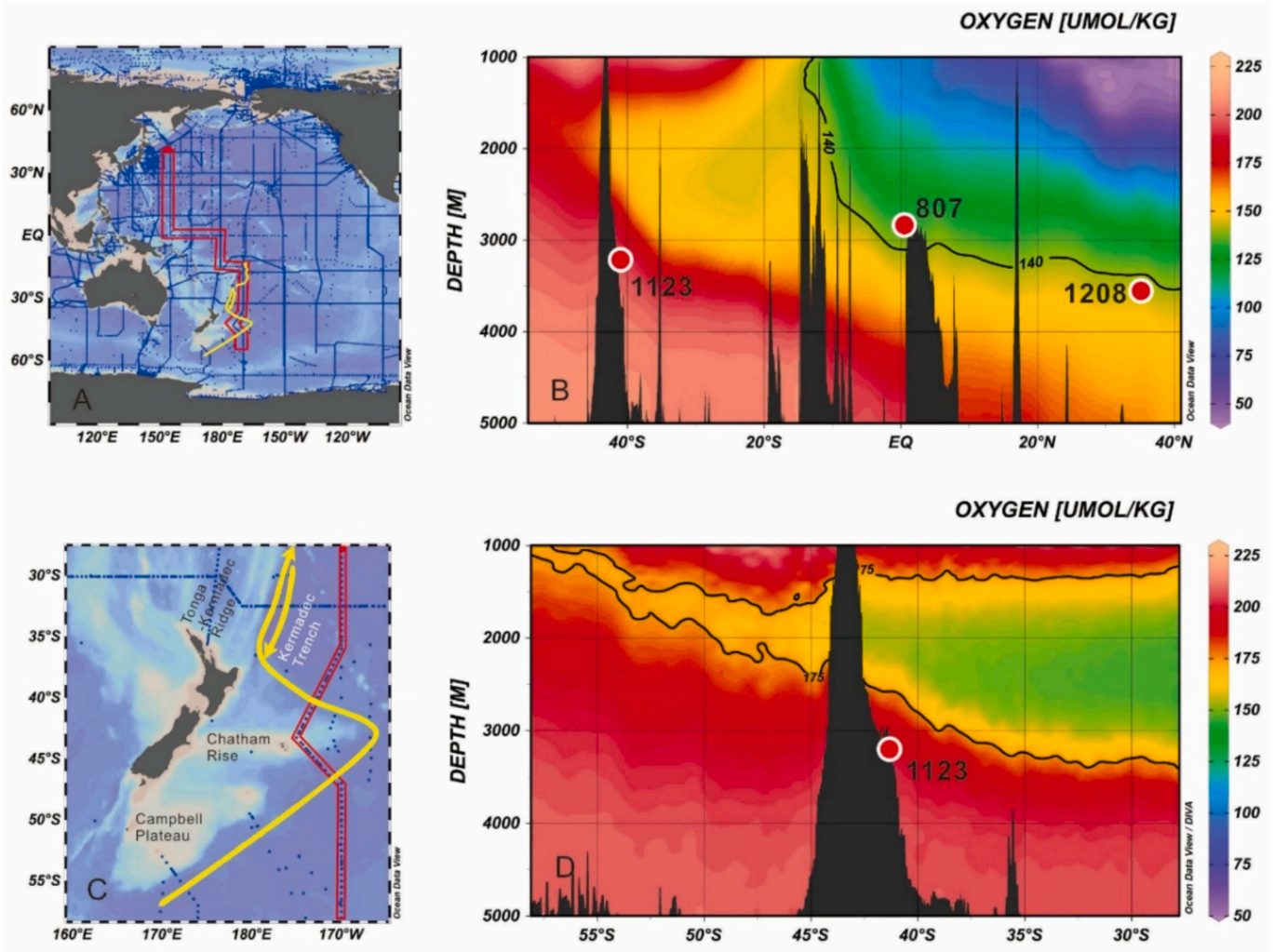


Fig. 2. Dissolved oxygen distribution along the Pacific DWBC (Lauvset et al., 2022). (A) Transect location; blue dots mark sampling stations (B) Dissolved oxygen concentration along the section. The black curve in (B) marks the 140 μmol kg^{-1} contour, separating PDW from LCDW (Talley, 2008). (C) Zoomed view of the South Pacific section. (D) Dissolved oxygen distribution near the Chatham Rise. The black curve in (D) indicates the 175 μmol kg^{-1} contour; oxygen levels below this threshold suggest UCDW influence (Talley, 2008). The yellow lines in A and C denote the modern DWBC (modified from Chandler et al. (2024)). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

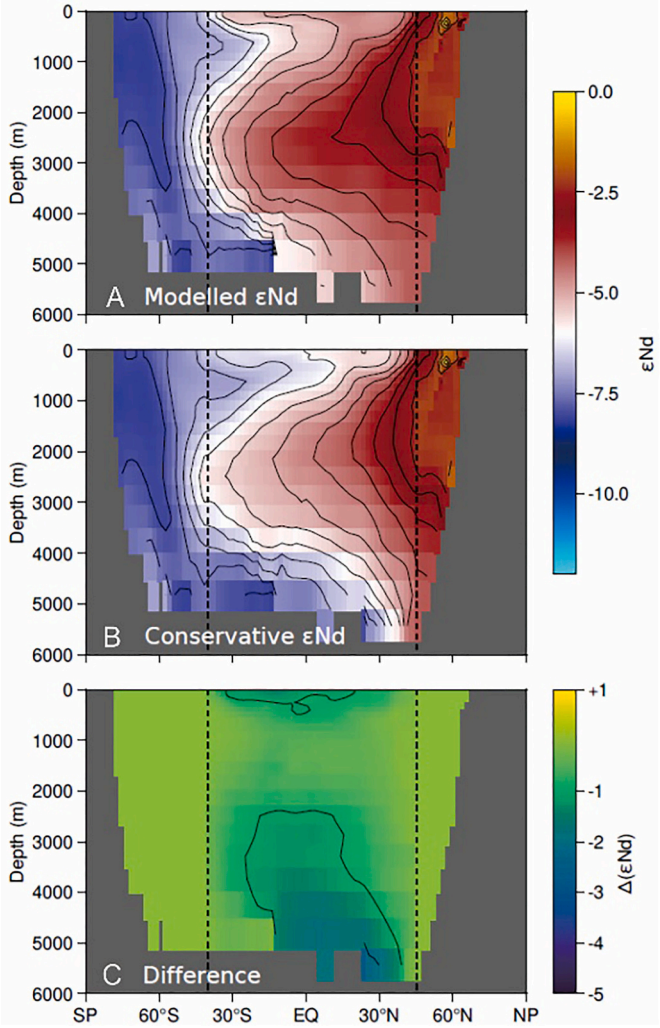


Fig. 3. Modeled influence of non-conservative processes on Pacific ε_{Nd} (Pasquier et al., 2022). The region between 40°S and 45°N is defined by dashed lines, where all non-conservative factors were removed to simulate purely mixing in panel (B). (A) Zonally averaged Pacific ε_{Nd} from full model output. (B) Zonally averaged Pacific ε_{Nd} from conservative mixing only. (C) Difference between (B) and (A), representing the effect of non-conservative processes on ε_{Nd} within the central Pacific.

southwardly, its ε_{Nd} gradually becomes unradiogenic because of mixing with UCDW and LCDW (Fig. 1B, Behrens et al., 2018). Above PDW lies the North Pacific Intermediate Water (NPIW), with maxima ε_{Nd} values at depths of 800–1000 m (−3.4 to −2.7, Amakawa et al., 2009).

In the modern, there is no deep-water formation in the high-latitude North Pacific, leading to limited surface water influences on the deep Pacific (Ferreira et al., 2018; Haug et al., 1999). However, during G-IG cycles, vertical shifts in water mass boundaries may likely occur (Herguera et al., 1992; Keigwin, 1982; Skinner et al., 2017), and large-scale meridional circulation may have strengthened (Hall et al., 2001; Hu and Piotrowski, 2018; Shankle et al., 2025).

Three sites from Ocean Drilling Program (ODP) along the DWBC are selected for this study. ODP Site 1123 (41°47′S, 171°30′W, 3290 m) is located on the Chatham Rise, east of New Zealand. It lies within LCDW, with a temperature of 1.34 °C and salinity of 34.72 PSU (Reagan et al., 2024). Its distance from land (>1000 km) limits terrestrial input, and low Australian dust flux minimizes dust deposition and reversible scavenging (Tanaka and Chiba, 2006). However, the core-top (Holocene archive) authigenic ε_{Nd} of ODP Site 1123 is 1.5–2 units more radiogenic than nearby seawater measurements (Fig. 5, Basak et al., 2015). Similar offsets at other sites (CHAT 1K/5K) on the Chatham Rise (Fig. 5) suggest a regional feature (Noble et al., 2013). Given that five vertical profiles (0–5000 m) in this region show systematic offsets in modern seawater ε_{Nd} among each other (Fig. 5, Basak et al., 2015), it is plausible that distinct seawater ε_{Nd} signatures exist within small distances. The sharp [O_2] gradient (Lauvset et al., 2022) above ODP Site 1123 (Fig. 2D) further suggests complex water mass geometry. These observations demonstrate that the offset between seawater and the core-top authigenic ε_{Nd} might reflect that they are locates at different water mass subunits. Records also indicate that during glacial periods, LCDW remained dominant at ODP Site 1123, but was likely more radiogenic due to greater PDW influence (Hu et al., 2016; Noble et al., 2013; Williams et al., 2024).

ODP Site 1208 (36°6′N, 158°12′E, 3346 m) is located on Shatsky Rise in the northwest Pacific, near the lower boundary of the PDW (Fig. 1 A/B). Modern bottom water bathing this site has a temperature of 1.51 °C, salinity of 34.667 PSU (Reagan et al., 2024), and ε_{Nd} of −3.5 (Fig. 6), confirming PDW dominance. Surface detrital sediments near ODP site 1208 have ε_{Nd} of −11 (Robinson et al., 2021), suggesting weak volcanic influence and a greater contribution from dust or continent sediment. Site 1208 lies along the main East Asian dust transport axis (Shao and Dong, 2006). Modern dust deposition is ~3–7 g m^{-2} $year^{-1}$ (Shao and Dong, 2006), with dust ε_{Nd} between −8 and −11 (Chen et al., 2007). Other continental sources have little impact, given the site's distance (~2000 km) from continent. However, it is located <1000 km from the Japan-Kuril volcanic arc with ε_{Nd} ranging from +6 to +8, which may exert a radiogenic imprint (GEOROC database, https://georoc.eu/). During glacial periods, a shift in the PDW/LCDW boundary may have increased LCDW presence at ODP site 1208 (Keigwin, 1982), leading to a more unradiogenic shift. Enhanced Asian dust deposition during glacials (Abell et al., 2021; Serno et al., 2015) likely influenced seawater ε_{Nd}.

ODP Site 807 (3°36′N, 156°38′E, 2807 m) is located on the Ontong-Java Plateau, within the southward-expanding PDW. Modern bottom water bathing this site has a temperature of 1.75 °C, salinity of 34.669 PSU (Reagan et al., 2024), and ε_{Nd} of −3 (Fig. 7). Surface detrital sediments near ODP site 807 have ε_{Nd} of −6 (Robinson et al., 2021). Lower [O_2] at site 807 compared to site 1208 (Fig. 2B) suggests that site 807 is influenced by older PDW along the DWBC flow path (Lauvset et al., 2022). The more radiogenic ε_{Nd} values at site 807 relatives to site 1208 also reflect the accumulation of radiogenic Nd along the PDW flow path. Reversible scavenging is thought to be limited at site 807 due to the low surface particle flux, although surface water ε_{Nd} is 4 units more radiogenic than bottom water (Grenier et al., 2013).

At ODP core 1208A, 65 samples were collected from 33.59 to 63.01 rmbsf (recompressed meters below the seafloor), covering 600.4–1300 ka (Ford and Raymo, 2019). At ODP core 807 A, 55 samples were taken from 8.86 to 19.03 rmbsf, spanning 600.9–1227.8 ka (Feng et al., 2022). At ODP cores 1123B and 1123C, 92 samples were obtained from 26.58 to 62.61 rmcd (revised meters composite depth), corresponding to 600–1300 ka (Elderfield et al., 2012). The average time resolution of the ε_{Nd} records is ~9.6 ka for site 1208, ~12 ka for site 807, and ~8.9 ka for site 1123, reasonably resolving ε_{Nd} variations on G-IG cycles. Increased temporal resolution was applied between 900 and 1100 ka, a key interval for ocean circulation changes during the MPT. To ensure age model unity, we conducted a comprehensive examination on previously established age models for ODP Sites 1208 (Ford and Raymo, 2019), 807 (Feng et al., 2022), and 1123 (Elderfield et al., 2012), based on high-resolution (1–3 kyr) benthic foraminiferal $\delta^{18}O$ records (Fig. 8).

2.2. ε_{Nd} analysis on FFT and bulk sediment

We carried out all chemical procedures and instrumental analysis at the State Key Laboratory of Marine Geology, Tongji University. For FFT ε_{Nd} analysis, sediment samples were oven-dried, soaked and washed through 63 μm sieves with deionized water. FFT were hand-picked from the >150 μm fractions under a microscope. They were sonicated in

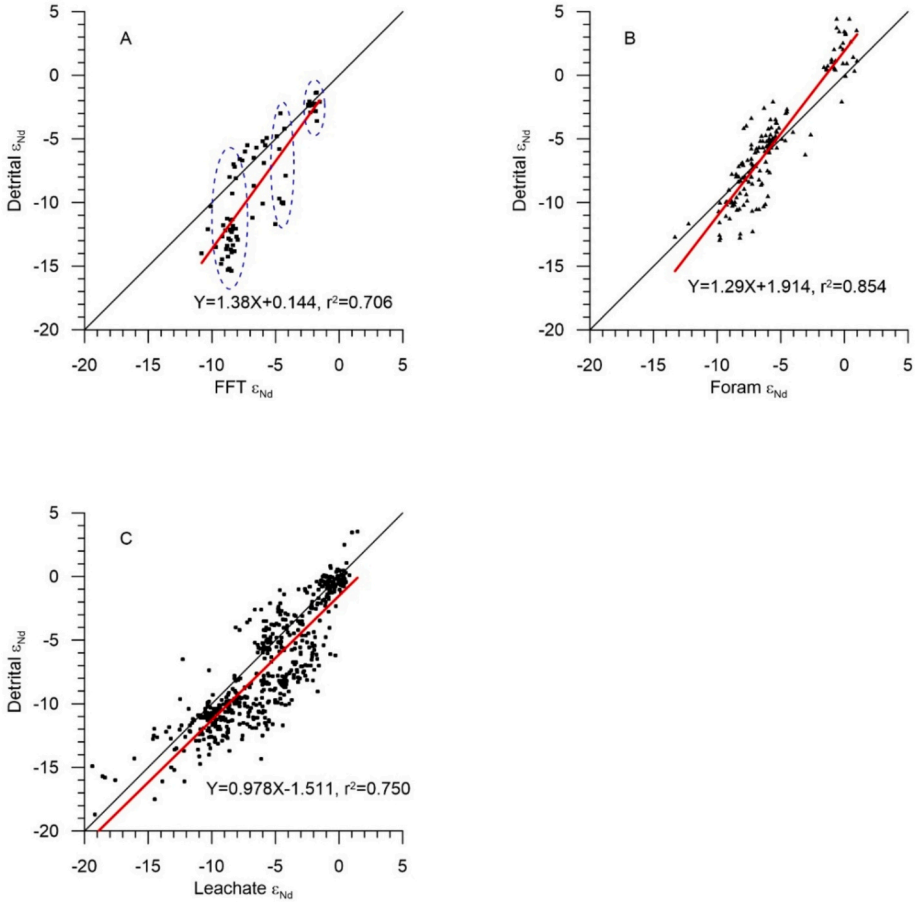


Fig. 4. Comparison of paired authigenic-detrital ε_{Nd} values. The figures are redrawn from Abbott et al. (2022), supplemented by additional data from Horikawa et al. (2021). (A) FFT, (B) Foraminifer, (C) Fe-Mn oxide leachates. Authigenic and detrital samples were taken from the same sediment samples. The black lines represent hypothetical 1:1 relationship between authigenic and detrital ε_{Nd} values, while the red lines show the linear regression of the observed data. A red line slope closer to 1 indicates stronger detrital influence. FFT show the largest deviation from the 1:1 line, indicating minimal detrital overprint. FFT's ε_{Nd} values also display distinct clusters (dashed ellipses), corresponding to major global water mass ε_{Nd} signatures, such as NPIW (−1.5; Horikawa et al., 2021), PDW (−4.5; McKinley et al., 2019) and CDW (−9; Scher et al., 2014; Martin et al., 2010). Relevant data are listed in Table S7. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

ultrapure water and methanol to remove clays, then subjected to reductive and oxidative treatments to remove Fe-Mn coatings and organic matters (Boyle and Keigwin, 1987). The cleaned FFT were dissolved in 6 N HCl. Before column chemistry, an aliquot of FFT sample was separated for trace element and REE concentration measurements. For bulk sediment analysis, 50 mg of dried and grounded samples were digested at 190 °C for 48 h in a 1:1 mixture of 16 N HNO_3 and 27 N HF. The solution was then evaporated to dryness, re-dissolved in 16 N HNO_3 and dry again. The residual was dissolved by 2 ml of 30 % HNO_3 and then heat at 190 ± 5 °C for 12 h. Nd was separated from both FFT and bulk sediment through a two-step protocol using Ion chromatography columns (Scher and Delaney, 2010; Wu et al., 2017). Eichrom TRU-spec resin was employed to isolate REEs, followed by Ln-spec resin for Nd separation. To ensure data quality, each batch of approximately 15 samples included a USGS reference material BHVO-2 (Weis et al., 2005). In addition, 22 duplicate FFT samples were analyzed.

ε_{Nd} were measured using a Thermo-Fisher Scientific Neptune-Plus Multiple Collector Inductively Coupled Plasma Mass Spectrometry (MC-ICP-MS) with an Aridus III desolvating nebulizer (CETEC Technologies) for high sensitivity. To ensure the accuracy and precision of the measurements, the reference standard JNdi-1 was meticulously run every five samples and for continuous monitoring external reproducibility. Instrumental mass discrimination was corrected relative to $^{146}Nd/^{144}Nd = 0.7219$ using an exponential law. All analyzed $^{143}Nd/^{144}Nd$ sample ratios were further normalized to 0.512115 of the JNdi-1 (Tanaka, 2000) to correct the machine bias.

2.3. GNOM v2.0

To constrain the effect of non-circulation factors on Pacific seawater ε_{Nd}, we adopted GNOM v2.0 for the global oceanic Nd cycle (Pasquier et al., 2022). The model is based on the steady-state circulation and employs a matrix representation to enable efficient computations, facilitating systematic parameter optimization. GNOM simulates six major Nd sources (Table S8), including benthic flux, atmospheric dust, volcanic ash, riverine input, groundwater discharge, and hydrothermal inputs, to characterize the distribution of Nd and ε_{Nd} in the ocean (Pasquier et al., 2022). The reversible scavenging scheme in GNOM has limitations, notably its omission of manganese oxide-mediated processes (Du et al., 2025). Despite this, the model reasonably captures observed [Nd] and ε_{Nd} distributions. Sensitivity experiments using GNOM remain valuable for diagnosing and constraining non-conservative Nd behavior in the ocean.

In this study, we modified the original "conservative diagnostic" code from Pasquier et al. (2022) to quantify the relative contribution of non-conservative processes to seawater ε_{Nd} in the Pacific (Fig. 3; https://github.com/Lee9251/GNOM-Pacific-Nd-isotope-experiments). We further used diagnostic tools provided by Pasquier et al. (2022) to conduct sensitivity experiments to evaluate the impact of various non-conservative mechanisms on Pacific ε_{Nd} (https://github.com/Lee9

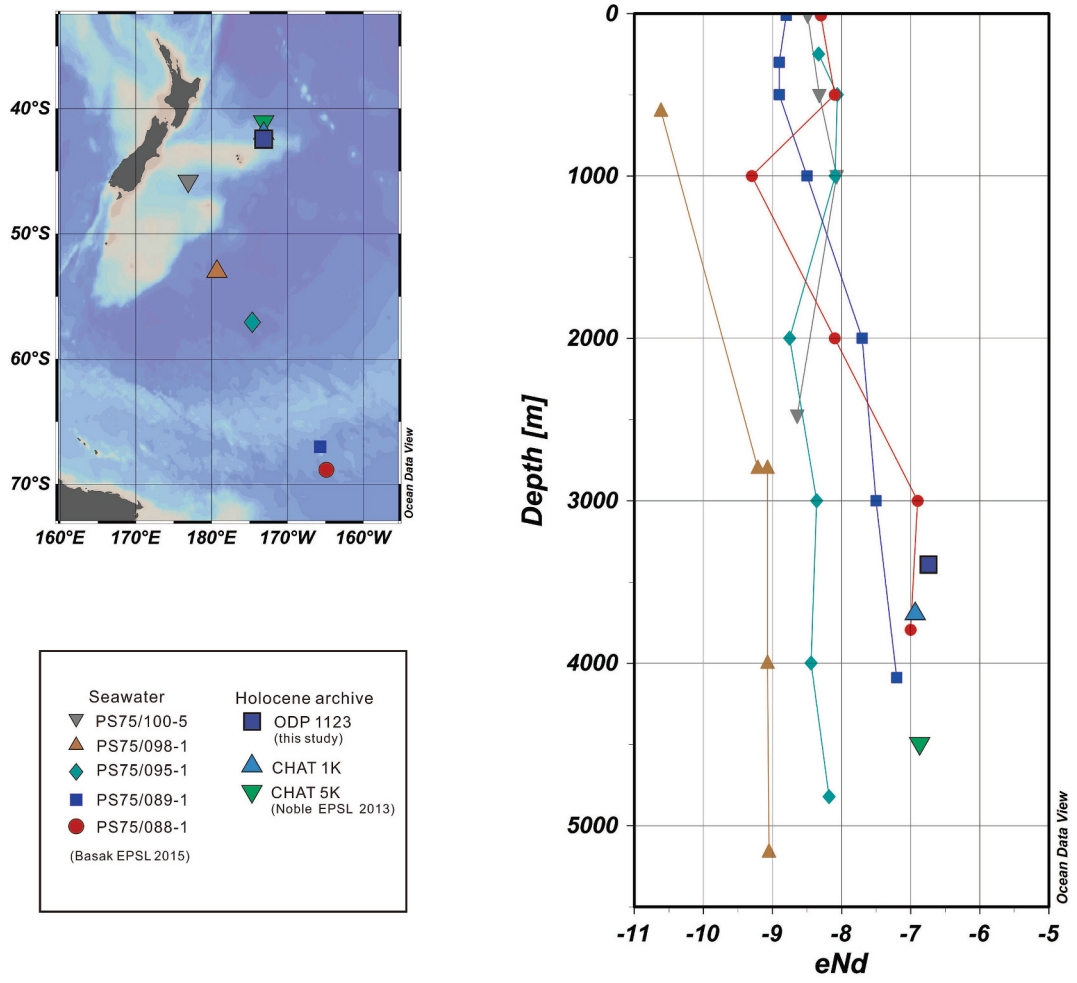


Fig. 5. Comparison of core-top authigenic ε_{Nd} and modern seawater ε_{Nd}. ODP Site 1123 (Basak et al., 2015; Noble et al., 2013).

251/GNOM-Pacific-Nd-isotope-experiments).

3. Results

Each ε_{Nd} record displays regular G-IG cycles, with relatively stable G-IG amplitude (Fig. 9). Site 807 ε_{Nd} remains continually more radiogenic than site 1208 ε_{Nd} by ~1 unit, displaying similar trend to the present. As reflected in the vertical profiles of the modern seawater ε_{Nd}, the bottom water ε_{Nd} at Site 1208 is around −4 (Fig. 6), which is less radiogenic than the bottom water ε_{Nd} (−3 to −3.5) at Site 807 (Fig. 7). The south Pacific site 1123 ε_{Nd} exhibits positive excursions during glacials but negative excursions during interglacials (Fig. 9D). However, ε_{Nd} values of the north Pacific sites 1208 and 807 show an opposite pattern (Fig. 9B/C). The opposite change of north and south Pacific deep water ε_{Nd} (Fig. 9B/C) leads to lower glacial and higher interglacial $\Delta\varepsilon_{Nd\,(807–1123)}$ values (Fig. 9E). The $\Delta\varepsilon_{Nd(807–1123)}$ exhibits a gradual decrease from MIS25 to MIS22 by ~3 ε (Fig. 9E), probably indicating a marked reorganization in Pacific deep circulation.

In deep-sea sediments, most Nd is in terrigenous material, with non-lithogenic material such as calcium carbonate contributing less than 10 %, so the bulk sediment ε_{Nd} can be used as a reference for detrital ε_{Nd} (Abbott et al., 2015; Du et al., 2016). The bulk sediment ε_{Nd} of site 1208 ranges from −8.5 to −12.2, substantially more negative than the FFT ε_{Nd} (−2.5 to −4.5, Fig. 10D). At this site, a radiogenic shift in bulk sediment ε_{Nd} occurs between 791 ka and 852 ka; however, no corresponding shift is observed in the FFT ε_{Nd}. At site 1123, bulk sediment ε_{Nd} generally ranges from −5.8 to −9.3, being lower than that of FFT (Fig. 10F), but with one extremely radiogenic value of ~1.36 occurring at ~1287 ka (Fig. 10F) that probably point to strong volcanic influence. However, the FFT ε_{Nd} of site 1123 displays no such a radiogenic shift at ~1287 ka (Fig. 10E), suggesting that it is a useful proxy to record past bottom water ε_{Nd} signals. The dataset including FFT and bulk sediment ε_{Nd} as well as the ε_{Nd} and $\delta^{13}C$ gradients was archived in Tables S1-S6.

To assess non-conservative processes influences, we conducted control run (modern, Fig. 11B) and 6 glacial sensitivity experiments by GNOM (Pasquier et al., 2022) and calculated seawater ε_{Nd} difference between control run and each sensitivity experiment (Fig. 11C-H). The results suggest that influence of non-conservative factors on the Pacific glacial seawater ε_{Nd} is limited. Indeed, a twofold increase in Asian dust flux yields a slight unradiogenic shift (~0.1 ε) in seawater ε_{Nd} (Fig. 11C). However, a concurrent enhancement of reversible scavenging counteracts this effect by introducing radiogenic surface signals into the deep ocean, ultimately producing a net radiogenic shift of up to ~0.4 ε (Fig. 11F/G). Increased volcanic activity produces minor radiogenic changes in surface waters but exerts minimal influence on deep waters (Fig. 11D). Increasing the volcanogenic component in sediments leads to ~0.2 ε shifts in deep water (Fig. 11E). Taking all these non-conservative factors into comprehensive consideration in sensitivity experiment (Fig. 11H) produces radiogenic shift of ~0.6 ε in deep waters, obviously conflicting with the reduced glacial $\Delta\varepsilon_{Nd(807–1123)}$ (Fig. 11E). The GNOM sensitivity experiments reveal that the non-conservative factors have limited influences on the G-IG variations of Pacific deep water ε_{Nd}.

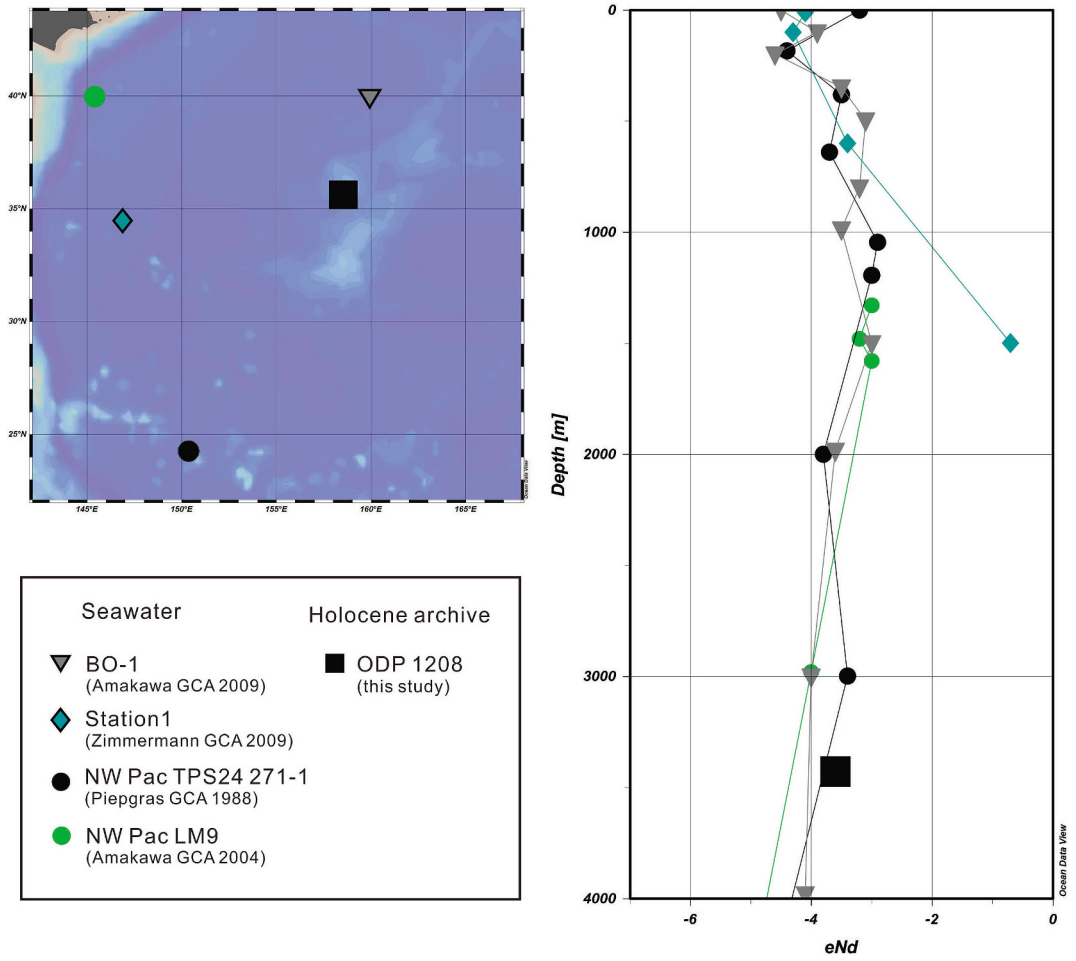


Fig. 6. Comparison of core-top authigenic ε_{Nd} and modern seawater ε_{Nd}. ODP Site 1208 (Amakawa et al., 2004; Amakawa et al., 2009; Piepgras and Jacobsen, 1988; Zimmermann et al., 2009).

4. Discussion

4.1. Constraining non-conservative influences on Pacific authigenic ε_{Nd}

Traditionally, variations in seawater ε_{Nd} have been primarily attributed to ocean circulation change (e.g. Amakawa et al., 2004; Hu and Piotrowski, 2018; Pena and Goldstein, 2014). Recent studies reveal the significant influence of non-conservative processes on seawater ε_{Nd} (Zhao et al., 2019), which underscores the necessity of constraining non-conservative influences before attributing glacial ε_{Nd} shifts to ocean circulation changes.

We focus on the $\Delta\varepsilon_{Nd(807–1123)}$ to restrict the factors influencing ε_{Nd} within the DWBC (Fig. 9E). At Site 1123, ε_{Nd} fluctuations align with those in the Atlantic (Hu and Piotrowski, 2018; Williams et al., 2024), more radiogenic during glacials, though with smaller amplitude. This may reflect changes in the ε_{Nd} composition of NADW (Zhao et al., 2019) or a reduced mixing proportion of NADW to the LCDW (Hu et al., 2016; Pena and Goldstein, 2014; Williams et al., 2024). Despite a potential offset between authigenic and seawater ε_{Nd} at this site (Fig. 5), we assume it remains stable over time, making $\Delta\varepsilon_{Nd(807–1123)}$ a representative indicator for the alteration of ε_{Nd} during seawater transport along the DWBC. Sites 807 and 1208 display broadly synchronous G-IG ε_{Nd} variability throughout the past 1.3 Ma, and their long-term offset closely matches the modern inter-site difference, indicating that both sites faithfully record large-scale deep Pacific circulation changes. However, site 1208 lies closer to the northern limit of SCW influence and is therefore more sensitive to modest fluctuations in local upwelling intensity, which can perturb its ε_{Nd} signal while leaving the equatorial site 807 largely unaffected. These localized effects likely account for the minor mismatches between sites 807 and 1208 during intervals such as MIS 21 and 26. Therefore, the $\Delta\varepsilon_{Nd(807–1123)}$ is better than $\Delta\varepsilon_{Nd(1208–1123)}$ in capturing the basin-scale deep circulation changes. We also examined the end-member gradient in ε_{Nd} between North Atlantic and Equatorial/North Pacific, but focused on the intra-Pacific gradient that better records the deep Pacific circulation changes.

To reduce the $\Delta\varepsilon_{Nd(807–1123)}$ during glacial periods (Fig. 9E), non-conservative processes must lead to more un-radiogenic ε_{Nd} at site 807. It is commonly believed that enhanced dust flux during glacials, particularly from Asia and Australia with more negative ε_{Nd} values, could lower Pacific ε_{Nd} (Amakawa et al., 2009; Du et al., 2020; Hu and Piotrowski, 2018). Indeed, increasing the Asian dust flux by a factor of two in GNOM results in a ~0.1-unit decrease in DWBC ε_{Nd} (Fig. 11C). However, dust also enhances reversible scavenging, introducing more radiogenic ε_{Nd} signals from surface water into the deep ocean. In GNOM, the combined effect of these mechanisms shifts deep water ε_{Nd} to more radiogenic values: with a 1.5× and 2× increase in dust flux and scavenging efficiency, DWBC ε_{Nd} rises by ~0.2 and ~0.4 units, respectively (Fig. 11F/G). Site 1208 is along the main dust transport axis (Shao and Dong, 2006), but it shows similar changes in FFT ε_{Nd} to that of site 807 (Fig. 9B/C), indicating that dust has a limited impact on the $\Delta\varepsilon_{Nd(807–1123)}$.

Volcanic activity is another important non-conservative factor influencing the Pacific seawater ε_{Nd}. While larger ice sheets during glacials may reduce volcanic activity by altering magma chamber

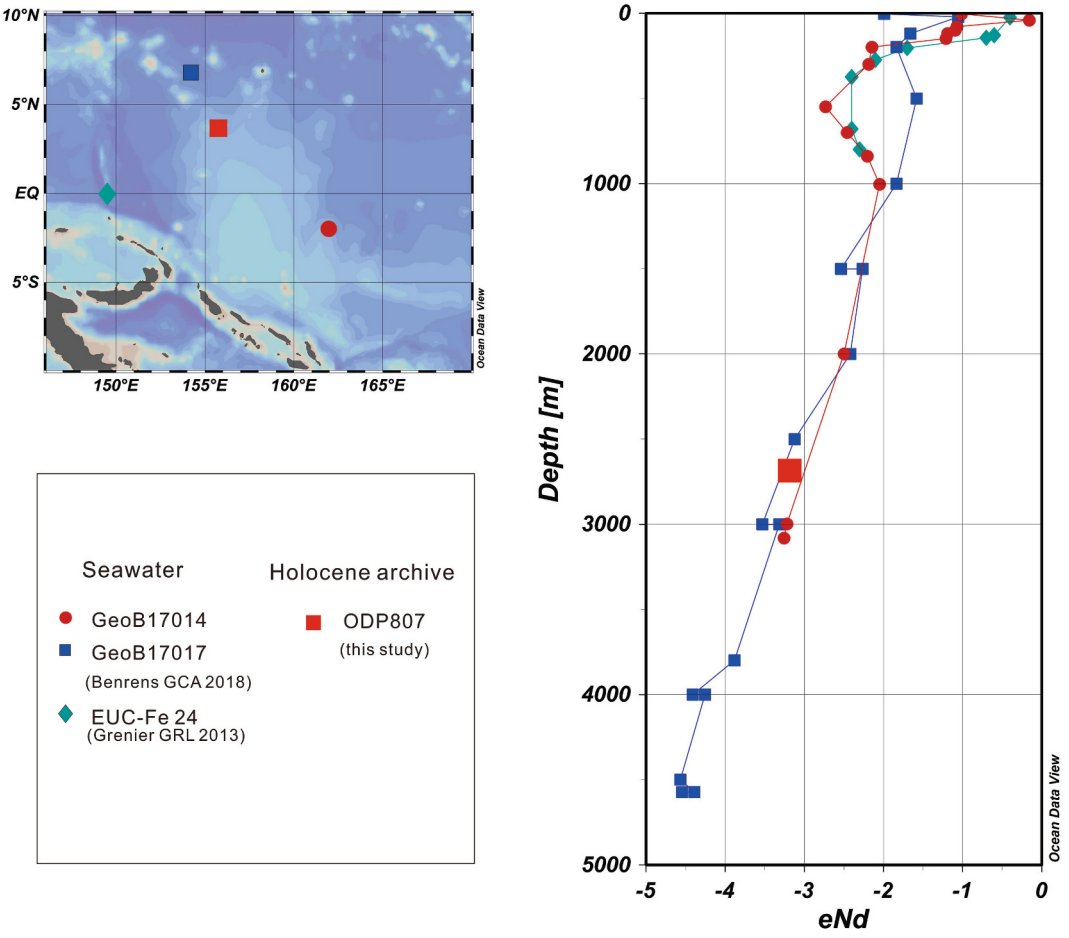


Fig. 7. Comparison of core-top authigenic ε_{Nd} and modern seawater ε_{Nd}. ODP Site 807 (Behrens et al., 2018; Grenier et al., 2013).

pressure (Huybers and Langmuir, 2009) or inhibiting the release of volcanic material (Du et al., 2018), any impact is generally restricted to high latitudes and thus has minimal effects on DWBC-related ε_{Nd} records. A lower sea level could enhance hydrothermal inputs from mid-ocean ridges (Huybers and Langmuir, 2017); but hydrothermal inputs account for only ~1 % of the global Nd flux (Pasquier et al., 2022), exerting limited influence. Lower sea levels could reduce the transport distance between volcanic sources and deep-sea, potentially increasing the delivery of volcanic material. Using GNOM simulations, we found that a 10 % increase of volcanic ash input only has little effect on the surface Pacific seawater ε_{Nd} (Fig. 11D), while increasing volcanic material content in all Pacific sediments leads to an increase of ~0.2 units in deep water ε_{Nd} (below 2000 m; Fig. 11E). The GNOM model results suggest that strengthened volcanic activity during glacials could increase deep water ε_{Nd} at site 807 and enlarge the $\Delta\varepsilon_{Nd(807-1123)}$, conflicting with the observed data. In summary, the glacial volcanic activity could not cause a marked decrease in ε_{Nd} at site 807.

The presence of benthic flux challenges the reliability of marine authigenic ε_{Nd} as a conservative proxy of ocean circulation (Du et al., 2025). Benthic flux refers to the continuous release of Nd from sediments and pore waters into the overlying bottom water, thereby modifying bottom water ε_{Nd} (Abbott et al., 2015). Its modification capacity depends on sediment properties, flux intensity, and bottom water residence time, among other factors. Du et al. (2020) found no significant G-IG changes in the basin-scale sediment provenance in the Pacific. Since there is no reliable proxy to represent the benthic flux strength on G-IG cycles, we adopt constant benthic flux on G-IG cycles during the Pleistocene. The influence of detrital diagenetic processes on authigenic ε_{Nd} exchanging via pore water (Abbott et al., 2022) is regarded as minor because our measured bulk sediment ε_{Nd} at three sites show marked differences from the FFT ε_{Nd} (Fig. 10).

In the southwest Pacific, the core top (Holocene archive) authigenic ε_{Nd} is consistent among site 1123 (this study and Williams et al., 2024) and two nearby cores CHAT 1K and 5K (Fig. 5, Noble et al., 2013). The Holocene archive ε_{Nd} is ~1.2–2 ε more radiogenic than the surface seawater ε_{Nd} but comparable to the deep seawater (3000–4000 m, LCDW) ε_{Nd} of three stations (PS75/088–1089–1 and 100–5) between 46°S and 69°S (Fig. 5, Basak et al., 2015). The five vertical profiles (0–5000 m) show systematic offsets in modern seawater ε_{Nd} among one another (Fig. 5, Basak et al., 2015), displaying localized ε_{Nd} variability of distinct sub-water masses within LCDW. Generally, we suggest that site 1123 FFT ε_{Nd} records LCDW variability on G-IG cycles, and the core top authigenic ε_{Nd} of sites 1208 and 807 is nearly the same as the nearby seawater ε_{Nd} (Figs. 6 and 7).

4.2. Pacific meridional deep ocean circulation changes during MPT

Besides the non-conservative factors, modern Pacific LCDW ε_{Nd} (ε_{Nd} = −8 to −9) is largely determined by mixing of two-endmembers, NADW (ε_{Nd} = −13.5) in the source and PDW (ε_{Nd} = −3.5), with limited contribution from AABW in the South Atlantic (ε_{Nd} = −9) and South Pacific (ε_{Nd} = −7) (Pena and Goldstein, 2014). Given unchanged NADW ε_{Nd} during the Pleistocene (Pena and Goldstein, 2014), reduced glacial ε_{Nd} of site 1123 could be explained by decreased contribution from NADW to CDW. It is noted that while the glacial Atlantic ε_{Nd} became more radiogenic after MIS 22 (Fig. 12F/G), which may reflect reduced weathering of unradiogenic lithologies under expanding ice sheets (Zhao et al., 2019), North Pacific ε_{Nd} remained relatively

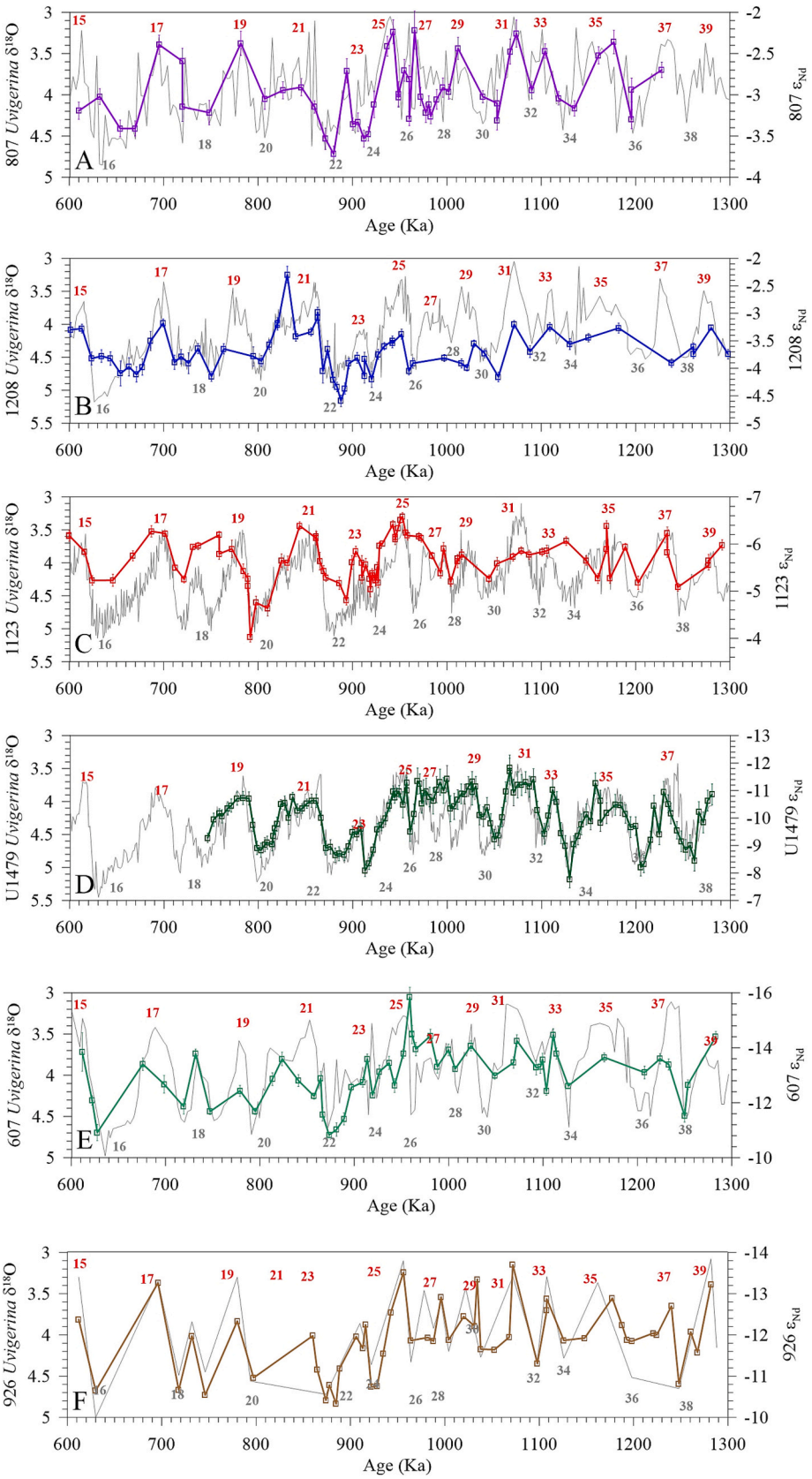


Fig. 8. Age model comparison. Grey lines are benthic foraminiferal $\delta^{18}O$ records (adjusted to *Uvigerina* spp.), and colored lines are ε_{Nd} records. (A) $\delta^{18}O$ (Feng et al., 2022) and ε_{Nd} (this study) for ODP 807; (B) $\delta^{18}O$ (Ford and Raymo, 2019) and ε_{Nd} (this study) for ODP 1208; (C) $\delta^{18}O$ (Elderfield et al., 2012) and ε_{Nd} (this study) for ODP 1123; (D) $\delta^{18}O$ (Hines et al., 2024) and ε_{Nd} (Hines et al., 2024) for IODP U1479; (E) $\delta^{18}O$ (Raymo et al., 1989) and ε_{Nd} (Kim et al., 2020) for ODP 607; (F) $\delta^{18}O$ (Bickert et al., 1997) and ε_{Nd} (Yehudai et al., 2021) for ODP 926.

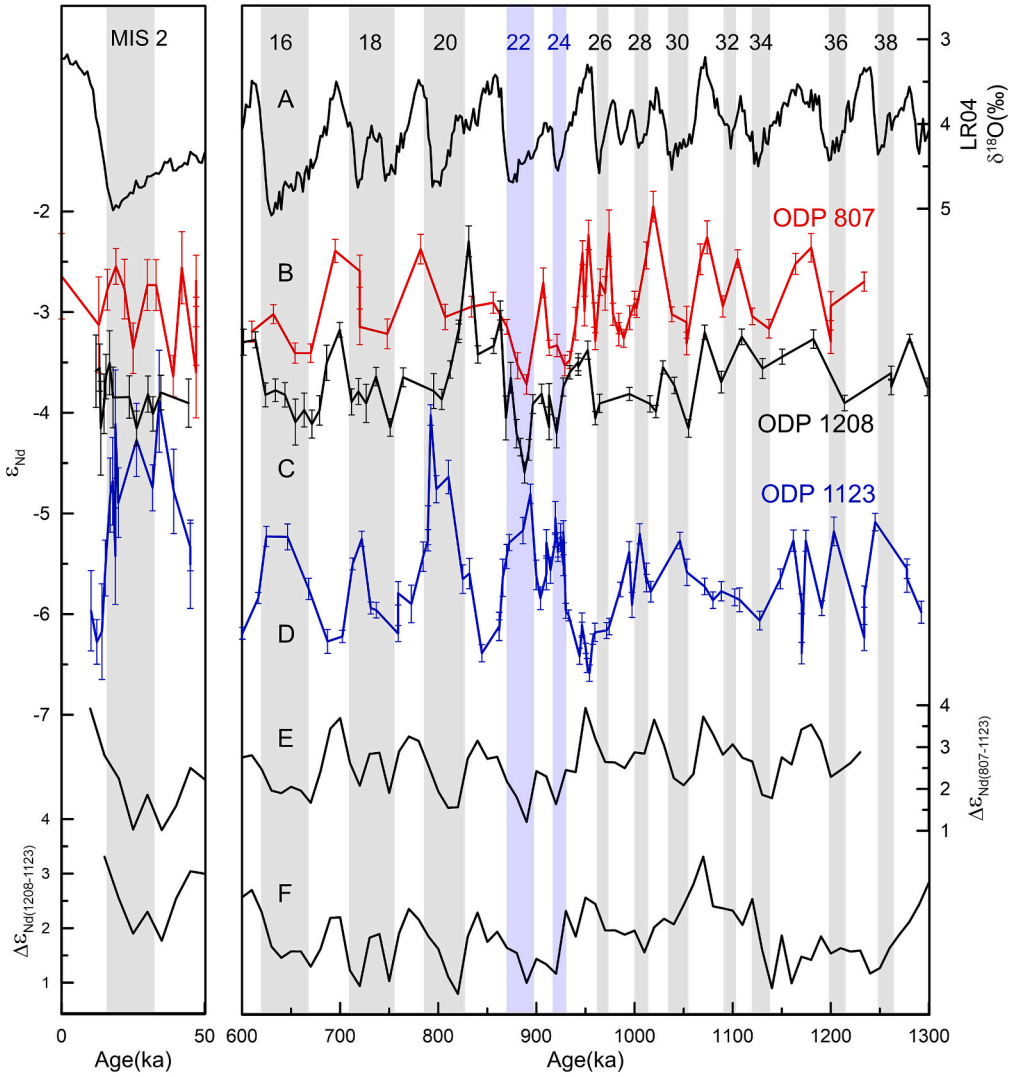


Fig. 9. (A) LR04 benthic foraminiferal $\delta^{18}O$ stack (Lisiecki and Raymo, 2005). Numbers indicate Marine Isotope Stages; glacial periods are shaded in grey, and MIS 22 and 24 are marked with blue bars. (B-D) ε_{Nd} records from ODP Sites 807, 1208, and 1123, respectively. (E) $\Delta\varepsilon_{Nd(807-1123)}$, ε_{Nd} gradient between sites 807 and 1123. (F) $\Delta\varepsilon_{Nd(1208-1123)}$, ε_{Nd} gradient between sites 1208 and 1123. Error bars represent $\pm 2\sigma$ internal analytical uncertainty. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

unchanged (Fig. 12B/C). This means that reduced glacial ε_{Nd} of site 1123 could also be caused by more radiogenic NADW during glacials due to a shift of water source to the Nordic Seas, as evidenced in LGM (Zhao et al., 2019). The glacial expansion of SCW could increase the contribution of LCDW in the north Pacific deep water and also shallow the PDW/LCDW interface (Ferrari et al., 2014), leading to reduced glacial deep water ε_{Nd} at sites 807 and 1208.

As LCDW flows northward, non-conservative processes and water mass mixing progressively enrich the water with radiogenic ε_{Nd} (Behrens et al., 2018). Meanwhile, the southward PDW continues to acquire radiogenic signatures via particle exchanges (Amakawa et al., 2004). In this case, faster flow would shorten the accumulation time for radiogenic modification, making the north Pacific water mass ε_{Nd} being less radiogenic. The ε_{Nd} gradient between the northern and southern deep Pacific, the $\Delta\varepsilon_{Nd(807-1123)}$, was principally influenced by 1) mixing with intermediate waters such as NPIW, 2) non-conservative Nd inputs, and 3) changes in deep-water ventilation and geometry. However, influences of both NPIW and non-conservative factors are thought to have been minimal or relatively stable during the MPT. Because PDW is derived predominantly from SCW, the observed $\Delta\varepsilon_{Nd(807-1123)}$ is therefore most plausibly interpreted as reflecting variations in SCW ventilation strength and/or changes in the PDW/LCDW water-mass boundary. Therefore, reduced glacial $\Delta\varepsilon_{Nd(807-1123)}$ (Fig. 12H) could be caused by decreased contribution of NADW to CDW, a vigorous Pacific DWBC and a shoaling of PDW/LCDW interface, but not a weakened NPIW formation that seems to be more vigorous during glacials (Rae et al., 2020).

Continuously accumulated respired carbon enriched in ^{12}C would make deep water more and more depleted in $\delta^{13}C$ (Schmittner et al., 2013) along the meridional flow path of DWBC, as evidenced in modern observation (Fig. 1C). This means that faster DWBC and shoaling of PDW/LCDW interface could increase the north Pacific deep water $\delta^{13}C$ (Schmittner et al., 2013) and thus increase the north-south Pacific deep water $\delta^{13}C$ gradient. This is confirmed in the $\Delta\delta^{13}C_{(807-1123)}$ record from 600 ka to 1300 ka, which show increased/decreased values during glacial/interglacial periods (Fig. 12I). Both deep water FFT ε_{Nd} and $\delta^{13}C$ records reveal faster DWBC and shoaling of PDW/LCDW interface in the glacial Pacific, supporting previous works based on sortable grainsize (Hall et al., 2001) and foraminiferal ε_{Nd} (Hu and Piotrowski, 2018) records in the southwest Pacific. Southern Ocean processes such as greater AABW production (Hu and Piotrowski, 2018) could cause the enhanced deep Pacific meridional ocean circulation, and so did a northward shift

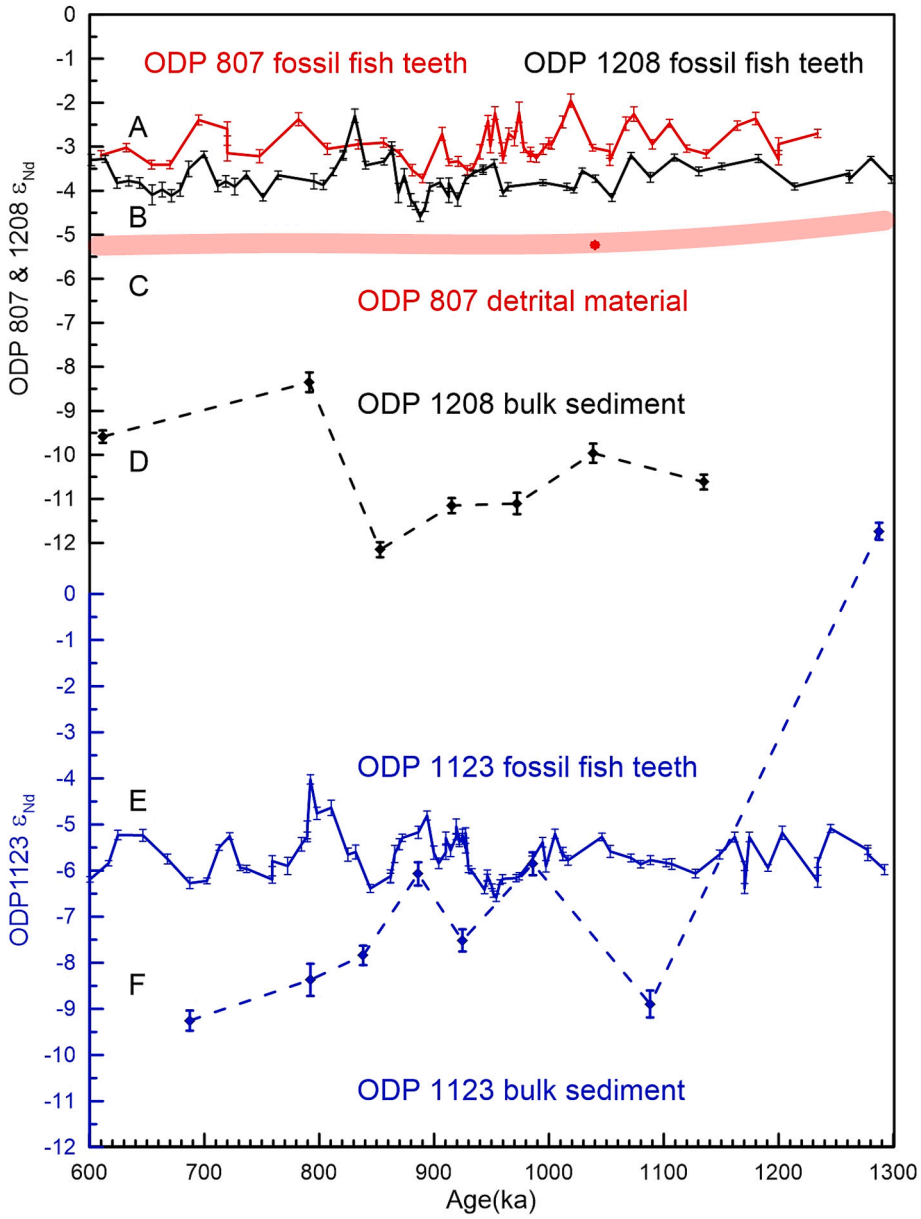


Fig. 10. Comparison of detrital and FFT ε_{Nd}. (A) FFT ε_{Nd} of ODP 807. (B) FFT ε_{Nd} of ODP 1208. (C) Detrital ε_{Nd} of ODP 807 (Le Houedec et al., 2016). (D) Bulk sediment ε_{Nd} of ODP 1208. (E) FFT ε_{Nd} of ODP 1123. (F) Bulk sediment ε_{Nd} of ODP 1123. Error bars ±2σ internal analytical uncertainty.

of the Southern Hemisphere westerly which could enhance Ekman transport in the Southern Ocean, thereby strengthening the ACC and promoting the CDW formation (Hall et al., 2001). The shoaling of the PDW/LCDW interface, as Ferrari et al. (2014) proposed, was potentially linked to expanded Southern Ocean sea ice, which may have displaced the isopycnals separating NADW/PDW from CDW upward and northward, leading to a vertical expansion of the lower overturning cell (LCDW). Previous studies based on $\delta^{18}O$ and $\delta^{13}C$ profiles in the North Pacific (Herguera et al., 1992; Keigwin, 1998) also discovered shoaling of the PDW/LCDW interface during the glacial period, supporting our interpretation on the ε_{Nd} data.

We also noticed the decoupling between the $\Delta\varepsilon_{Nd(807–1123)}$ and $\Delta\delta^{13}C_{(807–1123)}$ during the MPT. Due to influences from multi-factors, benthic foraminiferal $\delta^{13}C$ is less sensitive to ocean circulation change than seawater ε_{Nd} because these factors can mask circulation signals, whereas ε_{Nd} provides a more conservative constraint on water-mass mixing. This explains the divergence between $\Delta\varepsilon_{Nd(807–1123)}$ (Fig. 12H) and $\Delta\delta^{13}C_{(807–1123)}$ (Fig. 12I) across the MPT. Consequently, the muted $\Delta\delta^{13}C$ probably implies potential changes in deep ocean carbon storages after the MPT.

Overall, considering the relatively lower time resolution before MIS 24, the three Pacific FFT ε_{Nd} records exhibit no big differences on G-IG fluctuations (Fig. 12B/C/D), supporting modest oceanographic reorganization across the MPT (Hines et al., 2024). However, during MIS 22, the glacial ε_{Nd} of the north (807 and 1208) and south Pacific (1123) sites markedly shifted toward less radiogenic and more radiogenic values respectively, aligning with the synchronously more radiogenic shift in the Atlantic (Fig. 12E/F/G). A similar ε_{Nd} change with smaller-amplitude also occurred during MIS 24, marking the initial adjustment of Pacific deep circulation preceding MIS 22. The subsequent, larger excursion at MIS 22 represents the culmination of this transition. These patterns imply a transient circulation event of global significance, probably suggesting a worldwide abrupt expansion of SCW during MISs 24–22 (Pena and Goldstein, 2014). Overall, the Pacific ε_{Nd} reveals a post-MPT reorganization of deep circulation, with strengthened deep circulation contributing to a global thermohaline adjustment.

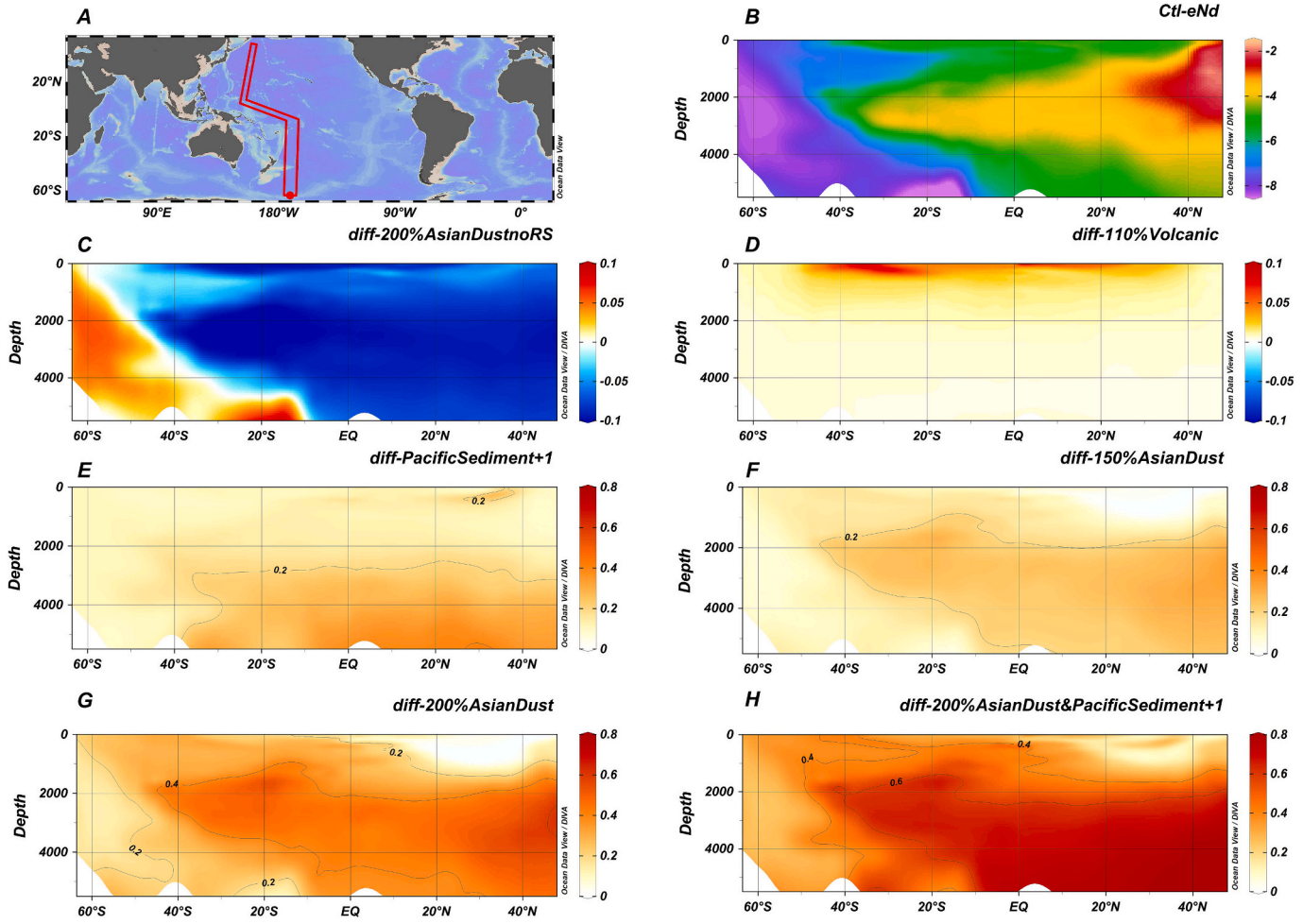


Fig. 11. Differences in seawater ε_{Nd} between control run experiment (modern) and 6 sensitivity experiments (glacial) by GNOM v2.0 model. (A) Red line represents Pacific meridional section, (B) control run experiment, using original parameters of Pasquier et al. (2022); (C) a twofold increase in Nd release from Asian dust, with constant reversible scavenging; (D) a 10 % increase in volcanic Nd flux; (E) an increase in volcanogenic component by adding +1 ε unit to all Pacific sediments; (F) a 1.5-fold increase in both Asian dust release and its reversible scavenging effect; (G) as in (F) but a twofold increase; and (H) as in (G), with an additional +1 ε unit added to Pacific sediments to simulate enhanced volcanic inputs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

4.3. Implication on ocean-carbon storage and climate evolution of the MPT

Previous work in the north-south Atlantic based on planktonic foraminiferal ε_{Nd} and epibenthic B/Ca demonstrate that increased deep ocean carbon storage associated with weakened Meridional Overturning Circulation (MOC) is intimately linked with the late Pleistocene 100-kyr G-IG cycles (Farmer et al., 2019). However, our ε_{Nd} records have revealed an acceleration of MOC during the glacial deep Pacific, which would reduce the respired carbon storage in the deep Pacific, not conducive to the glacial atmospheric pCO_2 decline. Referring to recent modelling work for the LGM conditions (Du and Tian, 2025), we argue for a different circulation-carbon relationship shaping the glacial Pacific deep ocean carbon storage. The modelling results demonstrated that synchronous cooling in the bi-polar oceans can substantially increase deep Pacific carbon storage, even under intensified overturning. This is attributed primarily to enhanced air-sea CO_2 disequilibrium in the Southern Ocean, which limits outgassing and promotes carbon retention. Additionally, brine rejection strengthened vertical stratification and further suppressing carbon release. Although the idealized experimentation based on the LGM conditions limits direct quantification for the MPT, the model results nonetheless establish a generic understanding for the Pacific carbon storage, that is, the trade-off between deep water transit time and surface disequilibrium. Indeed, the model simulations by Du and Tian (2025) that reveals co-existing of intensified overturning and increased deep-ocean carbon storage is in contrast to the conventional view that stronger ventilation necessarily reduces carbon sequestration. This model result is supported by our ε_{Nd} evidence. However, we also note that the ε_{Nd} proxy primarily constrains water-mass provenance and circulation changes rather than ocean carbon reservoir change; thus, the interpretation on carbon cycle based on ε_{Nd} records should be regarded as indirect evidences. The MPT marks the rise of the 100-kyr ice age cycle in the late Pleistocene. As demonstrated in the marine authigenic ε_{Nd} records in the Pacific and Atlantic oceans (Fig. 12), ocean circulation has played an important role in shaping the late Pleistocene ice ages through interaction with carbon cycle.

Our Pacific ε_{Nd} records indicate stable Pacific deep circulation on glacial cycles during the Pleistocene, which was punctuated by episodic intensification across the MIS 22. This intensification in deep Pacific circulation was probably triggered by strengthened northward penetration of the SCW, indicating that the Pacific actively participated in a global adjustment of the overturning circulation after the MPT and modulated the Pacific Ocean carbon reservoir change. As suggested by Clark et al. (2024), the increased Antarctic sea ice plays in enhancing formation of AABW and increasing ocean carbon storage during the MPT that caused an acceleration in the late Pleistocene global cooling and

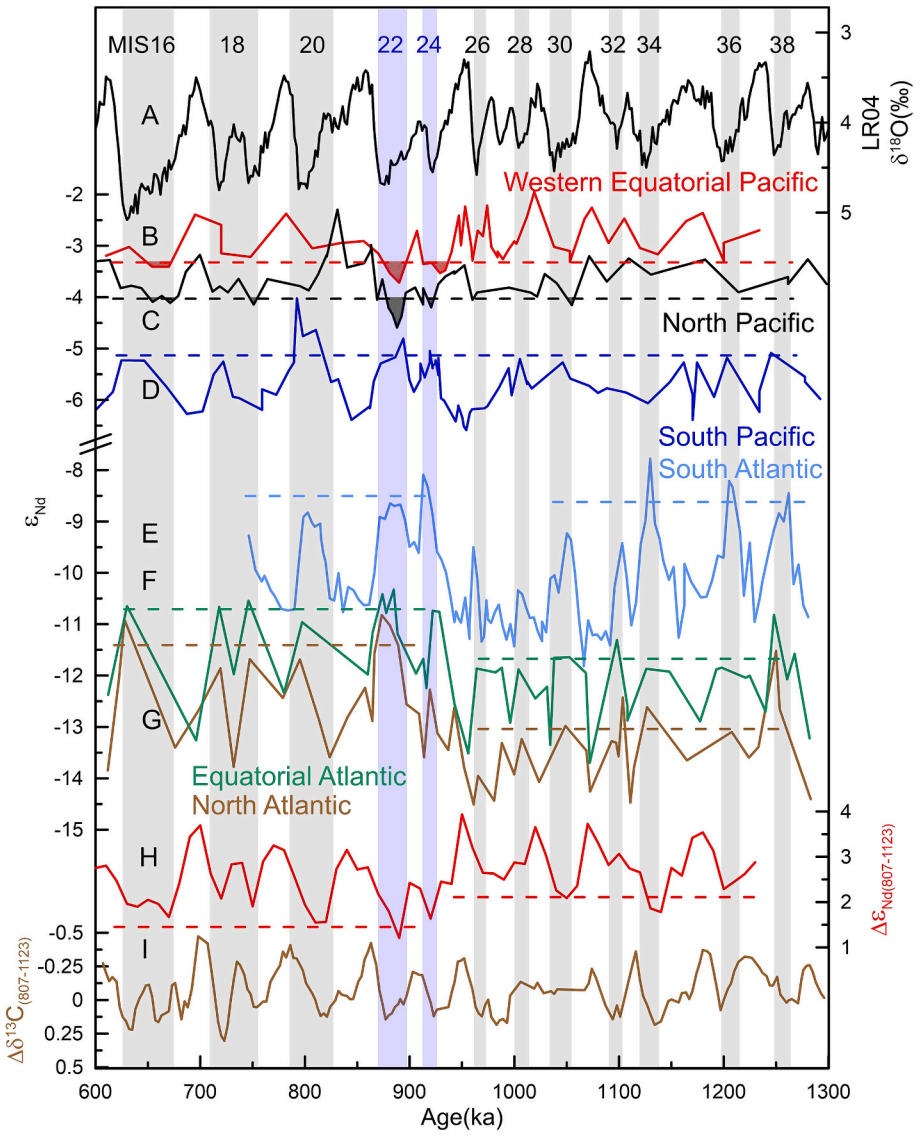


Fig. 12. (A) LR04 $\delta^{18}O$ (Lisiecki and Raymo, 2005). (B) ODP Site 807 ε_{Nd}. (C) ODP Site 1208 ε_{Nd}. (D) ODP Site 1123 ε_{Nd}. (E) IODP Site U1479 ε_{Nd} (Hines et al., 2024). (F) DSDP Site 926 ε_{Nd} (Yehudai et al., 2021). (G) ODP Site 607 ε_{Nd} (Kim et al., 2021). (H) $\Delta\varepsilon_{Nd(807-1123)}$, the ε_{Nd} gradient between ODP 807 and 1123. (I) $\Delta\delta^{13}C_{(807-1123)}$, the $\delta^{13}C$ gradient between ODP 807 and 1123, 5-points weighted average (Elderfield et al., 2012; Feng et al., 2022). Broken horizontal lines mark the mean glacial values for each time series, allowing comparison of the glacial baseline before and after the MPT.

promoted the significant 100-kyr glacial cycles during the post-MPT. Our records thus provide Pacific evidences of the marked changes in global ocean circulation and carbon cycle for this transition.

5. Conclusion

The north and south Pacific exhibit opposite changes in deep water ε_{Nd} during the MPT (600–1300 ka), which leads to reduced glacial ε_{Nd} gradient between north and south Pacific, probably resulted from enhanced expansion of SCW and/or shoaling of the PDW/LCDW interface. The MIS 22 within the MPT, an anomalously insolation-driven cold event (Elderfield et al., 2012), likely further reorganized the ocean circulation and enhanced ocean carbon storage, accelerating the progression of the late Pleistocene glaciation. Overall, the Pacific deep circulation underwent a marked reorganization after the MPT, and strengthened DWBC was part of a global adjustment of the thermohaline circulation. Enhanced glacial overturning and increased deep-ocean carbon storage are not contradictory; both were likely influenced by expanded Southern Ocean sea-ice cover after the MPT, and together contributed to the strengthening of late-Pleistocene glacial cycles.

CRediT authorship contribution statement

Ke Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation. **Jun Tian:** Writing – review & editing, Writing – original draft, Supervision, Investigation, Funding acquisition, Conceptualization. **Ning Zhao:** Writing – review & editing. **Jianghui Du:** Writing – review & editing. **Zhonghui Liu:** Writing – review & editing, Writing – original draft. **Jinlong Du:** Writing – review & editing, Writing – original draft. **Enqing Huang:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gloplacha.2025.105229.

Data availability

The authors confirm that all data necessary for supporting the scientific findings of this paper have been provided.

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